

#Task 1

```
text = "Artificial Intelligence"
```

```
print("First 10 characters:", text[:10])
print("Last 5 characters:", text[-5:])
print("Every 2nd character:", text[::2])
print("Reversed string:", text[::-1])
print("Extracted word:", text[11:])
```

```
===== RESTART: C:/Users/krsu/OneDrive/De
First 10 characters: Artificial
Last 5 characters: gence
Every 2nd character: Atfca nelgne
Reversed string: ecnegilletnI laicifitrA
Extracted word: Intelligence
```

#Task 2

```
email = "student@gmail.com"
```

```
username = email.split("@")[0]
domain_name = email.split("@")[1].split(".")[0]
domain_extension = email.split(".")[1]
```

```
print("Username:", username)
print("Domain name:", domain_name)
print("Domain extension:", domain_extension)
```

```
===== RESTART: C:/Users
Username: student
Domain name: gmail
Domain extension: com
```

#Task 3

```
numbers = [10, 20, 30, 40, 50]

numbers.append(60)
print("After adding 60:", numbers)

numbers.remove(30)
print("After removing 30:", numbers)

numbers.insert(2, 25)
print("After inserting 25 at index 2:", numbers)

print("Max:", max(numbers))
print("Min:", min(numbers))

numbers.reverse()
print("Reversed list:", numbers)
```

```
===== RESTART: C:/Users/krsb/OneDrive/Desktop/in
After adding 60: [10, 20, 30, 40, 50, 60]
After removing 30: [10, 20, 40, 50, 60]
After inserting 25 at index 2: [10, 20, 25, 40, 50, 60]
Max: 60
Min: 10
Reversed list: [60, 50, 40, 25, 20, 10]
```

#Task 4

```
data = [12, 45, 67, 23, 89, 45, 12, 90]

unique_data = list(set(data))
print("After removing duplicates:", unique_data)

unique_data.sort()
print("Ascending order:", unique_data)

unique_data.sort(reverse=True)
print("Descending order:", unique_data)
```

```
===== RESTART: C:/Users/krsb/OneDrive/Desktop/
After removing duplicates: [67, 12, 45, 23, 89, 90]
Ascending order: [12, 23, 45, 67, 89, 90]
Descending order: [90, 89, 67, 45, 23, 12]
```

>

#Task 5

```
t = (10, 20, 30, 40, 50)

print("First element:", t[0])
print("Last element:", t[-1])

print("Number of elements:", len(t))

print("30 exists:", 30 in t)

t_list = list(t)
print("Tuple converted to list:", t_list)
```

```
===== RESTART: C:/Users/krsb/OneDrive/I
First element: 10
Last element: 50
Number of elements: 5
30 exists: True
Tuple converted to list: [10, 20, 30, 40, 50]
```

#Task 6

```
student = {
    "name": "Rahul",
    "age": 21,
    "marks": 85
}

print("Keys:", student.keys())
print("Values:", student.values())

student["grade"] = "A"
print("After adding grade:", student)

student["marks"] = 90
print("After updating marks:", student)

student.pop("age")
print("After removing age:", student)
```

```
===== RESTART: C:/Users/krsb/OneDrive/Desktop/internship/review3.py ==
Keys: dict_keys(['name', 'age', 'marks'])
Values: dict_values(['Rahul', 21, 85])
After adding grade: {'name': 'Rahul', 'age': 21, 'marks': 85, 'grade': 'A'}
After updating marks: {'name': 'Rahul', 'age': 21, 'marks': 90, 'grade': 'A'}
After removing age: {'name': 'Rahul', 'marks': 90, 'grade': 'A'}
>>>
```

#Task 7

```
def add_numbers(a, b):
    return a + b

def check_even_odd(n):
    if n % 2 == 0:
        return "Even"
    else:
        return "Odd"

def factorial(n):
    fact = 1
    for i in range(1, n + 1):
        fact *= i
    return fact

print("Sum:", add_numbers(5, 3))
print("Even or Odd:", check_even_odd(7))
print("Factorial:", factorial(5))
```

```
===== RESTART: C:/Users/
Sum: 8
Even or Odd: Odd
Factorial: 120
|
```

#Task 8

```
for i in range(1, 6):
    for j in range(i):
        print(i, end="")
    print()
```

```
===== RESTART: C:/Use:
1
22
333
4444
55555
```

```
#Task 9
```

```
num = 12345
sum_digits = 0
n = num

while n > 0:
    sum_digits += n % 10
    n //= 10

print("Number:", num)
print("Sum of digits:", sum_digits)
```

```
===== RESTART: C:.\
Number: 12345
Sum of digits: 15
```

```
#Task 10
```

```
class Student:
    def __init__(self, name, age, marks):
        self.name = name
        self.age = age
        self.marks = marks

    def display(self):
        print("Name:", self.name)
        print("Age:", self.age)
        print("Marks:", self.marks)

s1 = Student("Subrahmanya", 22, 92)
s1.display()
```

```
===== RESTART: C:/User:
Name: Subrahmanya
Age: 22
Marks: 92
```

```
class BankAccount:
    def __init__(self, account_number, balance):
        self.account_number = account_number
        self.balance = balance

    def deposit(self, amount):
        self.balance += amount
        print("Deposited:", amount)

    def withdraw(self, amount):
        if amount <= self.balance:
            self.balance -= amount
            print("Withdrawn:", amount)
        else:
            print("Insufficient balance")

    def display_balance(self):
        print("Account Number:", self.account_number)
        print("Current Balance:", self.balance)

acc1 = BankAccount(123456, 5000)
acc1.deposit(2000)
acc1.withdraw(1500)
acc1.display_balance()
```

```
===== RESTART: C:/Users
Deposited: 2000
Withdrawn: 1500
Account Number: 123456
Current Balance: 5500
```

#Task 11

```
import pandas as pd
```

```
data = {  
    "Name": ["A", "B", "C"],  
    "Age": [20, 21, 19],  
    "Marks": [85, 90, 78]  
}
```

```
df = pd.DataFrame(data)
```

```
print("First row:")  
print(df.iloc[0])  
print("\nMarks column:")  
print(df["Marks"])
```

```
===== RESTART: C:/Users/ks
```

```
First row:
```

```
Name      A
```

```
Age       20
```

```
Marks     85
```

```
Name: 0, dtype: object
```

```
Marks column:
```

```
0      85
```

```
1      90
```

```
2      78
```

```
Name: Marks, dtype: int64
```

#Task 12

```
import pandas as pd
```

```
df = pd.read_csv("Students.csv")
```

```
print("First 5 rows:")  
print(df.head())
```

```
print("\nLast 5 rows:")  
print(df.tail())
```

```
print("\nColumn names:")  
print(df.columns)
```

```
df_no_null = df.dropna()  
print("\nData after removing null values:")  
print(df_no_null)
```

```
df_filled = df.fillna(df.mean(numeric_only=True))  
print("\nData after replacing null values with mean:")  
print(df_filled)
```

```
===== RESTART: C:/Users/krsb/OneDrive/Desktop/internship
```

```
First 5 rows:
```

	id	name	age	marks	city
0	1	Asha	21.0	85	Mysore
1	2	Ravi	22.0	90	Bangalore
2	3	John	NaN	78	Mangalore
3	4	Meena	21.0	abc	Delhi
4	5	Kiran	23.0	88	Chennai

```
Last 5 rows:
```

	id	name	age	marks	city
4	5	Kiran	23.0	88	Chennai
5	6	Ravi	22.0	90	Bangalore
6	7	Pooja	NaN	95	Mysore
7	8	Arun	24.0	70	Hyderabad
8	3	John	NaN	78	Mangalore

```
Column names:
```

```
Index(['id', 'name', 'age', 'marks', 'city'], dtype='object')
```

```
Data after removing null values:
```

	id	name	age	marks	city
0	1	Asha	21.0	85	Mysore
1	2	Ravi	22.0	90	Bangalore
3	4	Meena	21.0	abc	Delhi
4	5	Kiran	23.0	88	Chennai
5	6	Ravi	22.0	90	Bangalore
7	8	Arun	24.0	70	Hyderabad

```
Data after replacing null values with mean:
```

	id	name	age	marks	city
0	1	Asha	21.000000	85	Mysore
1	2	Ravi	22.000000	90	Bangalore
2	3	John	22.166667	78	Mangalore
3	4	Meena	21.000000	abc	Delhi
4	5	Kiran	23.000000	88	Chennai
5	6	Ravi	22.000000	90	Bangalore
6	7	Pooja	22.166667	95	Mysore
7	8	Arun	24.000000	70	Hyderabad
8	3	John	22.166667	78	Mangalore